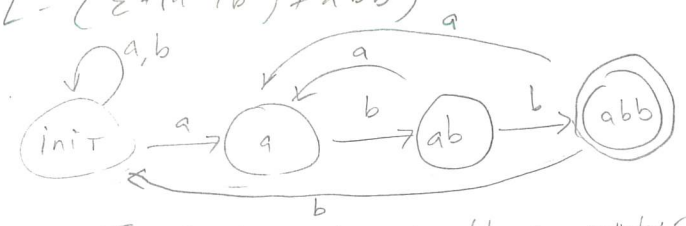
Chris Wolfe
Midterm 1 Theory

Ex 1

(Reg exp)

i) $L = \{ \varepsilon + (a^* + b^*)^* + abb \}$ X

ii)



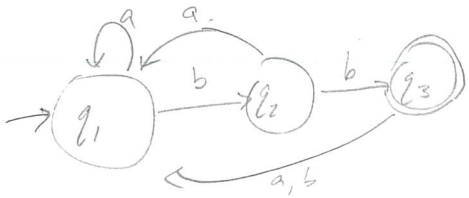
correct but overcomplicated.

Hand wavy at best

Justification: In order to get to state abb, any number of a's and b's must first occur, and any a might allow for the pattern abb which will be the accepting state.

iii) ?

iv)



No: bb is accepted here.

3 states - q_1, q_2, q_3

if the precursor to abb is b, it is in q_1 until it hits an a. then if there is a b, the only location to go is q_2. then if the final state is b, then the DFA can finish in the accepting state. if more characters are added, it must be re accepted by the DFA

Ex 2

i) $L = \{ \varepsilon + (b^* + a + b^* + a + b^* + a + b^*)^* \}$? Show your steps.

ii) ~~$\delta(q_i, q_i) = q_j$ where $j = (i + (i)_a) \bmod 3$~~

Base case $\delta(0, q_i) = q_j$ when $j = (i + \emptyset_a) \bmod 3$

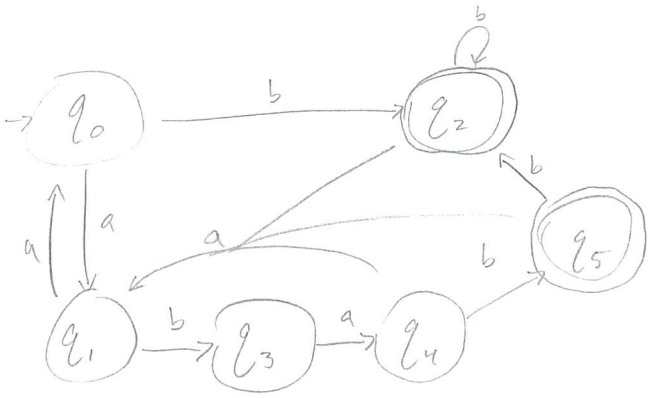
iii) The language accepts any word which has a multiple of 3 number of a's, interspersed with any number of b's (including $\emptyset$). ✓

Ex 3 on Reverse

Ex 3

Chris Wolfe

q_6 unreachable ✓



	q_0	q_1	q_2	q_3	q_4	q_5
q_0		1	✓	x	x	x
q_1			1	1	x	x
q_2				x	x	✓
q_3					1	x
q_4						✓
q_5						

where are the Os?